

ORIGINAL RESEARCH ARTICLE

Angiotensin Receptor Neprilysin Inhibitor in Heart Failure With Preserved Ejection Fraction and Secondary Mitral Regurgitation: The PRAISE-MR Randomized Trial

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BACKGROUND: Atrial functional mitral regurgitation (AFMR) characterizes a high-risk phenotype in heart failure with preserved ejection fraction (HFpEF). Although sacubitril/valsartan reduces functional mitral regurgitation (MR) in HF with reduced EF (HFrEF), its impact on exercise hemodynamics and the dynamic burden of AFMR in HFpEF remains to be elucidated.

METHODS: This multicenter, randomized, open-label trial with blinded primary endpoint assessment assigned 84 patients with symptomatic HFpEF and at least moderate AFMR within the previous year to sacubitril/valsartan (n=41) or standard-of-care (SOC; n=43). The primary outcome was the 6-month change in the exercise mean pulmonary arterial pressure to cardiac output (mPAP/CO) slope, assessed using cardiopulmonary exercise testing with simultaneous echocardiography (CPETecho). Secondary outcomes included changes in peak oxygen consumption (peak VO_2), Kansas City Cardiomyopathy Questionnaire (KCCQ), N-terminal pro-B-type natriuretic peptide (NT-proBNP) levels, left atrial (LA) volume and function, and AFMR severity in rest and during stress.

RESULTS: At 6 months, sacubitril/valsartan significantly improved the mPAP/CO slope compared with SOC (adjusted between-group difference in change, -0.93 mm Hg/L/min; 95% CI, -1.80 to -0.07 ; $P=0.035$). This hemodynamic benefit was accompanied by improvements in peak VO_2 (mean change, $+0.9$ versus -0.6 mL/kg/min; $P=0.002$) and KCCQ (median increase, 10 versus 2 points; $P=0.002$). Significant reductions in NT-proBNP and LA volume were observed ($P<0.001$ for both), alongside a significant blunting of the dynamic MR increase during exercise ($P=0.020$). Target dose was achieved in 60% of patients, with symptomatic hypotension as the primary titration-limiting factor.

CONCLUSIONS: In HFpEF and AFMR, sacubitril/valsartan was associated with improvements in exercise hemodynamics and peak VO_2 , along with attenuation of the exercise-induced increase in AFMR. These findings suggest a phenotype-specific benefit, warranting confirmation in larger, placebo-controlled, clinical outcome trials.

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Key Words: atrial function mitral regurgitation ■ heart failure with preserved ejection fraction ■ sacubitril/valsartan

Heat failure with preserved ejection fraction (HFpEF) has become the dominant form of heart failure globally, with its prevalence continuing to rise in parallel with an aging population and the increas-

ing burden of metabolic disorders.¹ Although the management of HFpEF has been fundamentally redefined, a substantial residual risk for hospitalization, mortality, and functional decline persists.^{2,3} This residual risk highlights

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Clinical Perspective

What Is New?

- This was the first randomized trial in atrial functional mitral regurgitation (AFMR) and heart failure with preserved ejection fraction (HFpEF) to demonstrate that sacubitril/valsartan significantly improves exercise hemodynamics by reducing the mean pulmonary arterial pressure to cardiac output (mPAP/CO slope).
- The treatment targets a dual mechanism, combining favorable structural reverse remodeling with the effective blunting of the dynamic increase in AFMR severity during physical exertion.
- The hemodynamic improvements were accompanied by significant gains in objective exercise capacity, patient-reported quality of life, and structural reverse remodeling.

What Are the Clinical Implications?

- Sacubitril/valsartan offers a phenotype-specific therapeutic strategy for the high-risk HFpEF population with AFMR.
- Pharmacological optimization should be prioritized to mitigate dynamic hemodynamic burden before considering invasive valve interventions in this population.
- Larger trials are required to determine whether these physiological improvements translate into reductions in mortality and heart failure hospitalizations.

the necessity for phenotyping to identify and treat specific pathophysiological drivers that remain unaddressed by current foundational therapies.

Secondary or functional mitral regurgitation (MR) identifies a high-risk phenotype, complicating 25% of the HFpEF population.^{4–7} Recognized as atrial functional mitral regurgitation (AFMR), this entity is driven by a deleterious interplay between atrial myopathy and annular dilation, a process frequently rooted in underlying left ventricular (LV) diastolic dysfunction.⁴ Because resting assessments often mask the complex pathophysiology of HFpEF and AFMR, stress-based characterization is often required to quantify their integrated hemodynamic burden.⁸ The mean pulmonary arterial pressure to cardiac output (mPAP/CO) slope serves as a flow-corrected measure of pulmonary resistance, integrating myocardial stiffness, vascular remodeling, and the hemodynamic impact of dynamic AFMR.^{9–11}

AFMR in HFpEF represents an unmet clinical need, as randomized trials to guide management are lacking.^{12,13} Although sacubitril/valsartan, an angiotensin receptor–neprilysin inhibitor (ARNI), reduces natriuretic peptide levels and left atrial (LA) volume in patients with HFpEF, and improves functional MR in HF with reduced EF (HFrEF), its capacity to specifically modify exercise hemodynamics and AFMR has not been defined.^{14–19}

Nonstandard Abbreviations and Acronyms

AFMR	atrial functional mitral regurgitation
ARNI	angiotensin receptor–neprilysin inhibitor
CPETecho	cardiopulmonary exercise testing with simultaneous echocardiography
EROA	effective regurgitant orifice area
HFpEF	heart failure with preserved ejection fraction
HFrEF	heart failure with reduced ejection fraction
KCCQ	Kansas City Cardiomyopathy Questionnaire
LARS	left atrial reservoir strain
LAVI	left atrial volume index
LMM	linear mixed-effect models
LV-GLS	left ventricular global longitudinal strain
mPAP/CO	mean pulmonary arterial pressure to cardiac output
MRA	mineralocorticoid receptor antagonist
peak VO₂	peak oxygen consumption
REML	restricted maximum likelihood
RVol	regurgitant volume
SGLT2i	sodium–glucose cotransporter 2 inhibitor
SOC	standard-of-care
VE/VCO₂	minute ventilation–carbon dioxide production

The phenotypic heterogeneity of HFpEF may underlie the inconsistent or modest treatment effects observed in previous sacubitril/valsartan trials.^{15,20} The PRAISE-MR trial (sacubitril/valsartan in heart failure with Preserved ejection fraction and Secondary Mitral valve Regurgitation) addresses this evidence gap by utilizing cardiopulmonary exercise testing with simultaneous imaging (CPETecho) to evaluate whether sacubitril/valsartan improves the hemodynamic profile and functional trajectory of patients with HFpEF and at least moderate AFMR.

METHODS

Design and Oversight

PRAISE-MR was a pragmatic, prospective, multicenter, randomized, open-label trial with blinded primary endpoint assessment. The protocol and statistical analysis plan were previously published (NCT05991284).²¹ Ethics approval was obtained from all centers and the Federal Agency for Medicines and Health Products (EudraCT: 2023-506634-70-00); all patients provided written informed consent. The study adhered to Good Clinical Practice and the Declaration of Helsinki, with oversight by a trial steering committee. The trial was conducted and reported in accordance with the CONSORT (Consolidated

Standards of Reporting Trials) guidelines. Data were managed via Castor EDC (Amsterdam, The Netherlands) and monitored by an independent team. Primary endpoints were adjudicated by a blinded, independent echocardiography core lab, with final statistical analyses verified by an independent statistician. Deidentified individual participant data will be made available upon reasonable request to the corresponding author.

Patient Inclusion

Symptomatic patients (NYHA class II–III) ≥ 40 years of age with HFpEF (LV ejection fraction [LVEF] $\geq 50\%$) and at least moderate AFMR (documented within 1 year) were eligible. HFpEF was diagnosed per ESC criteria and verified by the treating physician.²² AFMR was defined as at least moderate MR with normal LV dimensions and function, mitral annular and LA dilation, and loss of systolic leaflet concavity, without papillary muscle displacement. All patients received optimized background therapy before the baseline trial visit, including stable doses of sodium–glucose cotransporter 2 inhibitors (SGLT2i) and mineralocorticoid receptor antagonists (MRAs).^{22–24} Key exclusion criteria included systolic blood pressure < 100 mm Hg, estimated glomerular filtration rate ≤ 30 mL per minute per 1.73m^2 , potassium level > 5.4 mmol per liter, primary mitral valve disease, or inability to perform exercise testing.²¹ A comprehensive list of all inclusion and exclusion criteria is provided in Table S1.

Trial Procedures

Participants were randomized (1:1) to sacubitril/valsartan plus standard-of-care (SOC) or SOC alone via a web-based system, using permuted blocks of four and stratified by center (trial design in Figure S1). Sacubitril/valsartan was initiated at 24/26 mg bid and uptitrated every 2 weeks to a target dose of 97/103 mg bid, as tolerated. Angiotensin-converting enzyme (ACE) inhibitors were discontinued ≥ 36 hours before sacubitril/valsartan initiation, whereas angiotensin receptor blockers (ARBs) were switched directly. In both groups, 24-hour ambulatory blood-pressure monitoring guided antihypertensive optimization (target $< 130/80$ mm Hg) at baseline and 6 months. In the SOC group, this was achieved by titrating baseline medications or adding antihypertensive agents per current guidelines.²⁵ For patients with atrial fibrillation (AF), rhythm management decisions were required before study inclusion. CPETech was performed at baseline and 6 months; the acquisition protocol and mPAP/CO slope derivation have been previously described in detail and is briefly outlined in the Supplemental Material.^{8,9,21,26} Despite trivial TR, contrast enhancement with agitated colloid ensures high feasibility rate for reliable, multi-stage mPAP/CO slope calculations, a method well-validated against invasive measures.²⁷ At each site, CPETech studies were performed by a single investigator. Deidentified images, blinded to randomization, were analyzed by a central core laboratory for the primary endpoint and MR quantification with direct data entry into an electronic case report form (Castor EDC).

Outcomes

The primary outcome was the change from baseline to six months in the mean pulmonary arterial pressure to cardiac output (mPAP/CO) slope, as assessed by CPETech. Secondary

outcomes, assessed as the change from baseline to 6 months, included AFMR severity at rest and during exercise; LA volume and function; N-terminal pro-B-type natriuretic peptide (NT-proBNP) levels; KCCQ (scores range from 0 to 100, with higher scores indicating fewer symptoms and physical limitations), and CPET parameters, specifically peak oxygen consumption (pVO_2) and the minute ventilation–carbon dioxide production (V_E/V_{CO_2}) slope. Safety was evaluated through daily surveillance by the study team for prespecified adverse events of interest, including hospitalization, hypotension, renal failure, hyperkalemia, and angioedema.

Statistical Analysis

Baseline characteristics were compared using *t* tests, Mann-Whitney *U*, or Fisher's exact tests, as appropriate. Efficacy and safety analyses followed intention-to-treat and as-treated principles, respectively. The initial sample size of 110 patients was estimated to provide 80% power to detect a 0.5 mm Hg/L/min difference in the mPAP/CO slope (2-sided $\alpha=0.05$; 10% dropout rate). As PRAISE-MR is the first randomized trial to utilize the mPAP/CO slope as a primary endpoint, a conservative difference of 0.5 mm Hg/L/min was prespecified as the minimum clinically relevant effect size for the sample size calculation. Recruitment concluded at 84 randomized patients. Although theoretical power for the initial conservative difference of 0.5 mm Hg/L/min was 58% in this final cohort, the trial remained robustly powered for the larger observed biological effect (-0.93 mm Hg/L/min), yielding a post hoc power $> 95\%$. The primary endpoint (mPAP/CO slope at 6 months) was analyzed using an ANCOVA model. In this model, the 6-month follow-up mPAP/CO slope was the dependent variable, with the baseline mPAP/CO slope included as a fixed-effect covariate to provide unbiased treatment effect estimates. Treatment assignment was included as a fixed factor. For the primary multivariable adjustment, age, BMI, AF, and sex were added as additional fixed-effect covariates. Secondary outcomes were evaluated using linear mixed-effects models in a long format, incorporating all available measurements at both baseline and six months. These models included random intercepts for each participant to account for within-subject correlation. Fixed effects were included for treatment assignment, time, and the treatment-by-time interaction, with the latter serving as the primary estimator of the treatment effect. Categorical outcomes were analyzed using generalized linear mixed-effects models following the same longitudinal protocol. NT-proBNP was log-transformed to address right-skewness for the primary model analysis (LLM). To ensure clinical interpretability, descriptive statistics are reported as medians with their corresponding interquartile ranges. The treatment effect was back-transformed and expressed as the ratio of geometric means with its corresponding 95% CI. Model assumptions were verified via Q-Q and residual plots. Missing primary outcome data ($n=8$; 5 withdrawals, 3 technical failures) were handled via complete-case analysis for the ANCOVA model, as prespecified. For secondary outcomes, missing data were addressed under the Missing At Random assumption using linear mixed-effect models (LMM) with restricted maximum likelihood (REML) estimation, which utilizes all available longitudinal observations. The primary outcome was further evaluated through a sensitivity analysis using a linear mixed-effect model to account for all longitudinal

observations under the missing-at-random assumption. All tests were two-sided ($\alpha=0.05$). Analyses were performed using IBM SPSS Statistics version 22 (IBM Corp., Chicago, IL).

RESULTS

Patient Characteristics

From February 29, 2024, through September 24, 2025, a total of 84 patients were enrolled at 2 centers in Belgium. Of these, 41 patients were randomly assigned to receive sacubitril/valsartan and 43 were assigned to receive SOC (Figure 1). Although the original enrollment target was 110 patients, recruitment was concluded at 84 because of slower-than-anticipated inclusion rate. Five patients (6%) withdrew from the trial (3 sacubitril/valsartan and 2 SOC), below the projected 10% dropout rate, because of locomotor impairment ($n=1$) or refusal of follow-up evaluation ($n=4$). Among the 79 patients who completed the study, the mPAP/CO slope could not be obtained in 3 cases because of insufficient image quality. All patients remained in their randomly assigned treatment group throughout the study. In the SOC group, 20 patients (47%) were treated with either an ACE inhibitor ($n=10$) or an ARB ($n=10$). The 2 groups were balanced at baseline with respect to demographic and clinical characteristics, echocardiographic findings, and exercise performance (Tables 1–3).

mPAP/CO Slope

At baseline, median mPAP/CO slope was 3.6 (2.4–4.4) mm Hg/L/min in the SOC group and 3.9 (3.0–5.8) mm Hg/L/min in the intervention group. At the 6-month follow-up, the median mPAP/CO slope was 3.9 (2.9–4.8) mm

Table 1. Characteristics of the Patients at Baseline

	Standard-of-care (n=43)	Sacubitril/ valsartan (n=41)	P value
Age (y)	73±7	75±5	0.122
Women (%)	29 (67)	31 (75)	0.407
BMI (kg/m ²)	27±5	27±5	0.737
BSA (m ²)	1.85±0.19	1.82±0.20	0.537
Heart rate (bpm)	65±12	66±13	0.840
Systolic BP (mm Hg)	145±14	142±16	0.315
Diastolic BP (mm Hg)	78±9	80±11	0.211
NYHA class (%)			0.026
II	33 (77)	22 (54)	
III	10 (23)	19 (46)	
Atrial fibrillation (%)	27 (63)	34 (82)	0.068
Hypertension (%)	36 (84)	34 (83)	0.922
Diabetes mellitus (%)	8 (19)	5 (12)	0.417
Smoking history (%)	20 (47)	24 (59)	0.407
Dyslipidemia (%)	30 (70)	25 (61)	0.397
Coronary artery disease (%)	7 (16)	11 (27)	0.239
Peripheral artery disease (%)	3 (7)	5 (12)	0.415
OSAS (%)	4 (9)	4 (10)	0.944
Pacemaker (%)	9 (21)	3 (7)	0.075
MRA (%)	41 (95)	38 (93)	0.548
SGLT2i (%)	41 (95)	40 (98)	0.236
Loop diuretic (%)	10 (23)	10 (24)	0.993
Beta-blocker (%)	30 (70)	34 (83)	0.273
ACE/ARB (%)	20 (47)	13 (32)	0.373
Statin (%)	28 (65)	23 (56)	0.398
KCCQ	81 (57–87)	66 (49–85)	0.115
Hemoglobin (g/dL)	14±1	14±1	0.581
eGFR (mL/min/1.73m ²)	58±16	57±14	0.449
NT-proBNP (ng/L)	468 (270–904)	575 (380–1068)	0.157

ACE indicates angiotensin-converting enzyme; ARB, angiotensin receptor blocker; BMI, body mass index; BP, blood pressure; bpm, beats per minute; BSA, body surface area; eGFR, estimated glomerular filtration rate; MRA, mineralocorticoid receptor antagonist; n, number of patients; NT-proBNP, N-terminal pro-B-type natriuretic peptide; NYHA, New York Heart Association; OSAS, obstructive sleep apnea syndrome; and SGLT2i, sodium-glucose cotransporter 2 inhibitor.

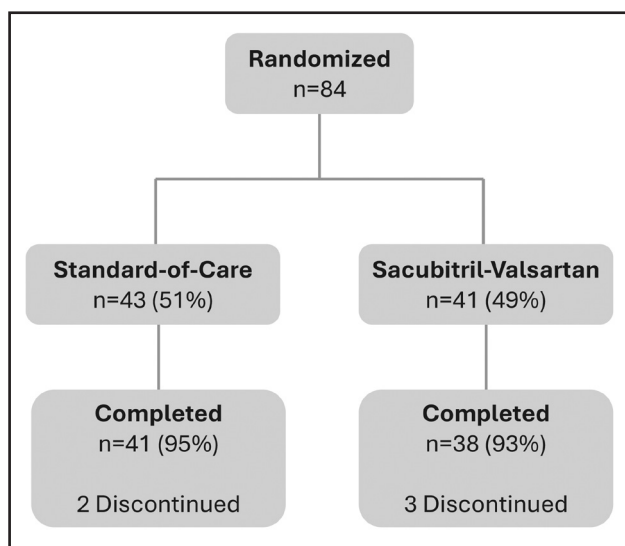


Figure 1. Participant flow diagram.

Flow of participants through the trial, including randomization to standard-of-care or sacubitril-valsartan, and completion rates at follow-up.

Hg/L/min in the SOC group and 3.7 (2.4–6.0) mm Hg/L/min in the intervention group. Overall, the intervention led to a significant improvement in the mPAP/CO slope compared with the SOC. The adjusted mean difference between the intervention and SOC groups was -0.93 mm Hg/L/min (95% CI, -1.80 to -0.07 ; $P=0.035$; Figure 2). The treatment effect remained consistent in a multivariable sensitivity analysis adjusting for age, sex, presence of AF, and BMI (adjusted between-group difference, -0.91 mm Hg/L/min; 95% CI, -1.80 to -0.02 ; $P=0.045$). This benefit was further supported by a higher categorical response rate, as

Table 2. Baseline Resting Cardiac Structure and Function

	Standard-of-care (n=43)	Sacubitril/valsartan (n=41)	P value
LV septum (mm)	10.6±1.8	10.9±2.4	0.553
LV EDD (mm)	49±6	47±5	0.057
LV EF (%)	55±5	54±5	0.656
LVEDV (mL)	96 (74–111)	93 (83–105)	0.566
LVESV (mL)	42 (33–53)	42 (37–50)	0.809
LV GLS (%)	15±4	14±3	0.316
Mitral annulus (mm)	37±4	37±5	0.687
LA diameter (mm)	38 (35–42)	40 (35–45)	0.300
LAVI (mL/m ²)	42±14	44±22	0.565
LA reservoir strain (%)	14 (9–18)	12 (9–16)	0.173
Mitral E wave (cm/s)	73±22	74±23	0.780
Mitral A wave (cm/s)	58±23	54±25	0.473
Septal e' (cm/s)	6 (6–7)	6 (5–8)	0.310
LVOT diameter (cm)	2 (1.9–2.2)	2 (1.9–2.1)	0.660
TR gradient (mm Hg)	24±5	22±6	0.314
TAPSE (mm)	18±4	17±5	0.293
RV FAC (%)	50±8	47±8	0.185
TR (%)			0.935
None	29 (67)	29 (71)	
Mild	12 (28)	10 (24)	
Moderate	2 (5)	2 (5)	
MR (%)			0.943
None	3 (7)	4 (10)	
Mild	22 (51)	22 (54)	
Moderate	17 (40)	14 (34)	
Severe	1 (2)	1 (2)	
MR EROA (cm ²) Feasible in 67 (80%)	0.17 (0.09–0.26)	0.14 (0.07–0.22)	0.222
MR regurgitant volume (mL) Feasible in 67 (80%)	24 (9–41)	22 (12–39)	0.180
MR vena contracta (mm) feasible in 70 (83%)	3 (2–4)	3 (2–4)	0.471

EDD indicates end-diastolic diameter; EDV, end-diastolic volume; EF, ejection fraction; EROA, effective regurgitant orifice area; ESV, end-systolic volume; FAC, fractional area change; GLS, global longitudinal strain; LA, left atrial; LAVI, left atrial volume index; LV, left ventricular; LVOT, left ventricular outflow tract; MR, mitral regurgitation; n, number of patients; RV, right ventricle; TAPSE, tricuspid annular plane systolic excursion; TR, tricuspid regurgitation; and vti, velocity-time integral.

70% of patients in the intervention group achieved an absolute reduction in mPAP/CO slope versus 36% in the SOC group ($P=0.003$). Sensitivity analysis using a linear mixed-effect model corroborated these findings, demonstrating a significant treatment-by-time interaction ($P=0.005$) and a net treatment effect of -1.30 mm Hg/L/min.

Mitral Regurgitation

Multiparametric resting AFMR severity significantly improved across the total population over 6-month follow-up (MR grade $P=0.003$; effective regurgitant orifice area [EROA] $P=0.045$; regurgitant volume [RVol] $P=0.040$; Figure 3), with comparable changes ob-

served between treatment groups (P interactions 0.973, 0.232, and 0.356, respectively; Table S2). Although exercise triggered an overall MR increase ($p=0.002$ for EROA), sacubitril/valsartan significantly attenuated this exercise-induced increase compared with SOC. Between-group comparisons adjusting for baseline values demonstrated that treatment with sacubitril/valsartan resulted in a significantly lower MR burden at mid-exercise for both EROA ($P=0.020$; Figure 4; Table S2) and RVol ($P=0.026$). Within the sacubitril/valsartan group (intragroup analyses), mid-exercise EROA and RVol improved significantly from baseline ($P=0.017$ and $P=0.029$, respectively), whereas no significant changes were observed in the SOC group ($P=0.938$ and $P=0.567$, respectively).

Table 3. CPETecho Characteristics of the Patients at Baseline

	Standard-of-care (n=43)	Sacubitril/valsartan (n=41)	P value
Cardiopulmonary exercise testing			
Forced vital capacity (L) % predicted	2.86±0.91 92±20	2.91±0.83 99±19	0.797 0.092
Forced expiratory volume (L) % predicted	2.15±0.70 91±22	2.14±0.64 96±21	0.976 0.239
Resting diffusing capacity (% predicted)	90±26	86±19	0.565
Peak heart rate (bpm)	106±23	108±26	0.803
Peak power (watt)	68±27	61±28	0.254
Peak VO ₂ (mL) % predicted	15.2±3.6 83 (71–95)	14.2±3.9 77 (66–92)	0.200 0.218
Ve/VCO ₂ slope	34 (29–37)	33 (30–40)	0.522
End-tidal peak CO ₂ (mm Hg)	34 (30–36)	33 (29–36)	0.626
Oxygen uptake efficiency slope (L/min/log10)	1.42 (1.19–1.76)	1.38 (1.09–1.58)	0.281
Peak RER	1.03±0.06	1.01±0.07	0.061
Peak systolic BP (mm Hg)	178±24	181±27	0.556
Peak diastolic BP (mm Hg)	81±13	85±17	0.208
Stress Echocardiography			
Peak TR gradient (mm Hg)	46±9	48±10	0.861
Peak TAPSE (mm)	24±5	22±5	0.173
Peak MAPSE (mm)	17±3	17±3	0.985
Peak septal e' (cm/s)	9 (7–11)	10 (8–10)	0.301
mPAP/CO slope (mm Hg/L/min)	3.5 (2.5–4.4)	3.9 (3.0–5.8)	0.052
Mid-exercise E/e'	13 (10–16)	12 (9–14)	0.059

BP indicates blood pressure; bpm, beats per minute; CO, cardiac output; E/e', ratio of inflow velocity to early diastolic velocity of the mitral annulus; e', early diastolic velocity of the mitral annulus; FEV, forced expiratory volume; FVC, forced vital capacity; MAPSE, mitral annular plane systolic excursion; mPAP, mean pulmonary arterial pressure; n, number of patients; OUES, oxygen uptake efficiency slope; RER, respiratory exchange ratio; TAPSE, tricuspid annular plane systolic excursion; TR, tricuspid regurgitation; VE/VCO₂, ventilatory equivalent for carbon dioxide; and VO₂, oxygen uptake.

Health Status and Exercise Capacity

Sacubitril/valsartan significantly improved health status and objective exercise capacity in addition to SOC. At 6 months, median KCCQ increased by 10 (3–15) versus 2 (–1–9) points (*P* interaction=0.002; Figure 5; Figure S2). Concurrently, peak VO₂ improved by 0.9±1.7 in the sacubitril/valsartan group versus –0.6±2.2 mL/kg/min in SOC; *P* interaction=0.002 (Tables S3 and S4). Ventilatory efficiency, assessed by the V_E/V_{CO2} slope, remained comparable (*P* interaction=0.296).

Biomarkers and Cardiac Remodeling

Sacubitril/valsartan led to a significant reduction in NT-proBNP levels. The ratio of change in geometric mean NT-proBNP was 0.49 (95% CI, 0.36 to 0.66; *P*<0.001) in the sacubitril/valsartan group compared with the SOC group, representing a 51.5% greater reduction (Figure 5; Figure S2). Further echocardiographic assessment showed evidence of structural and functional reverse remodeling. Compared with SOC, the intervention significantly reduced LAVI (mean change, –3.6±6.7 versus 2.3±4.9 mL/m²; *P* interaction<0.001; Tables S3 and S4)

and improved myocardial deformation, as measured by LV-GLS (mean change, –0.5±2.3 versus 0.9±1.7%; *P* interaction=0.003). LA reservoir strain (LARS) remained stable and comparable between groups (mean change, –0.1±5% intervention versus +0.1±4.0% in SOC; *P* interaction=0.814).

Safety

Among patients receiving sacubitril/valsartan (n=38), target (97/103 mg), intermediate (49/51 mg), and low doses (24/26 mg) were achieved in 60%, 24%, and 16%, respectively. Titration was limited in 15 patients by symptomatic hypotension (n=12) or asymptomatic protocol-defined criteria (n=3; hyperkalemia, hypotension, or 1.5-fold creatinine increase, n=1 each). Estimated glomerular filtration rate (eGFR) decreased more in the intervention group (interaction *P*=0.008; mean change –4±8 versus +1±7 mL/min/1.73m²; Figure S3), whereas potassium levels remained comparable (interaction *P*=0.294).

At 6 months, systolic blood pressure values were comparable (118±14 versus 121±9 mm Hg; *P*=0.255; Figure S4), whereas mean diastolic blood pressure was lower in the intervention arm (68±9 versus 73±6 mm

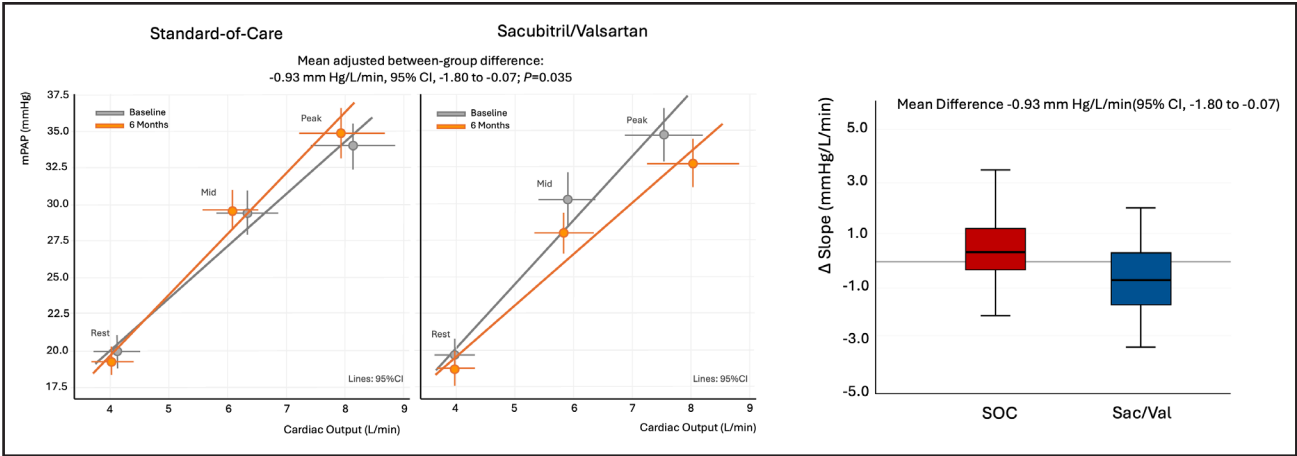


Figure 2. Impact of sacubitril/valsartan on the pulmonary pressure-flow relationship during exercise from baseline to 6-month follow-up. Mid indicates measurement taken during submaximal (intermediate) exercise; mm Hg, millimeters of mercury; mPAP, mean pulmonary arterial pressure; Sac/Val, sacubitril/valsartan; SOC, standard-of-care; and Δ, delta (change).

Hg; $P=0.033$; Figure S4). However, over 6 months, no significant treatment-by-time interaction was observed for systolic (P interaction=0.710) or diastolic (P interaction=0.095) 24-hour ambulatory blood pressure. At 2 weeks, antihypertensive therapy was up-titrated in 3 SOC patients, whereas background antihypertensive therapy was down-titrated in 4 intervention patients. Three CV-related hospitalizations occurred (one intervention and 2 SOC), with no mortality or de novo AF.

DISCUSSION

The PRAISE-MR trial is the first prospective randomized trial to demonstrate that sacubitril/valsartan significantly improves exercise hemodynamics, functional capacity, and quality of life in patients with HFpEF and AFMR. The

central finding is that sacubitril/valsartan treatment resulted in a significant reduction of the mPAP/CO slope, which was accompanied by structural reverse remodeling and a blunting of the dynamic MR increase during exercise.

AFMR is a frequent and high-risk complication within the HFpEF population, with its prevalence increasing to as high as 50% in the presence of concomitant AF.^{4–8} This underscores a deleterious bidirectional relationship: HFpEF or AF trigger LA remodeling and annular dilation, leading to AFMR, whereas the resulting volume overload further exacerbates atrial myopathy and disease progression. Notably, even mild AFMR can exert significant hemodynamic effects in the setting of a noncompliant LA.^{8,28} The PRAISE-MR trial required patients to have at least moderate AFMR documented within the previous year. However, the initiation of contemporary background therapies, including SGLT2i, MRAs, and rhythm management, often led to regression of AFMR severity at rest (>50% of cases) before study baseline, a phenomenon that likely reflects the dynamic and treatable nature of this atrial phenotype.^{29–31}

The primary endpoint, the mPAP/CO slope, serves as a flow-corrected marker of total pulmonary resistance, thereby capturing the interplay between myocardial stiffness, pulmonary vascular remodeling, and the volume overload of dynamic MR.^{9,10,32,33} The observed reduction of 0.93 mm Hg/L/min under sacubitril/valsartan likely reflects a synergistic dual mechanism. First, sacubitril/valsartan was linked to LA reverse remodeling and improved LV mechanics with parallel decrease in NT-proBNP values over time. This mirrors findings that neprilysin inhibition enhances myocardial load handling through mechanisms beyond blood pressure control that remain to be fully elucidated.^{14,16,17} Second, sacubitril/valsartan blunts the dynamic MR increase during exercise according to our trial. This is relevant, as even a modest

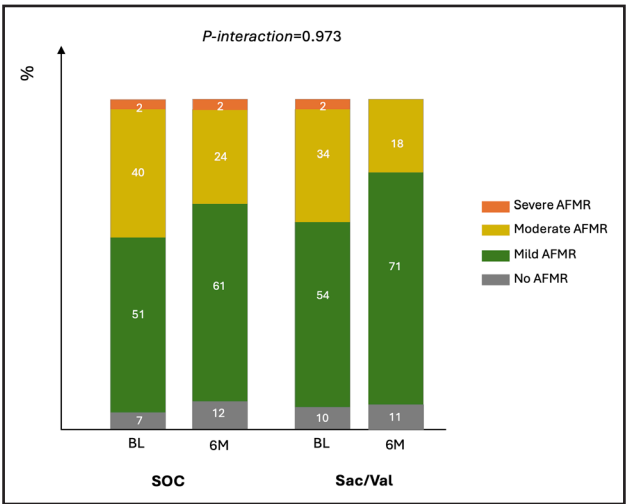


Figure 3. Changes in resting atrial functional mitral regurgitation severity over time. 6M indicates 6-month follow-up; AFMR, atrial functional mitral regurgitation; BL, baseline; Sac/Val, sacubitril/valsartan; and SOC, standard-of-care.

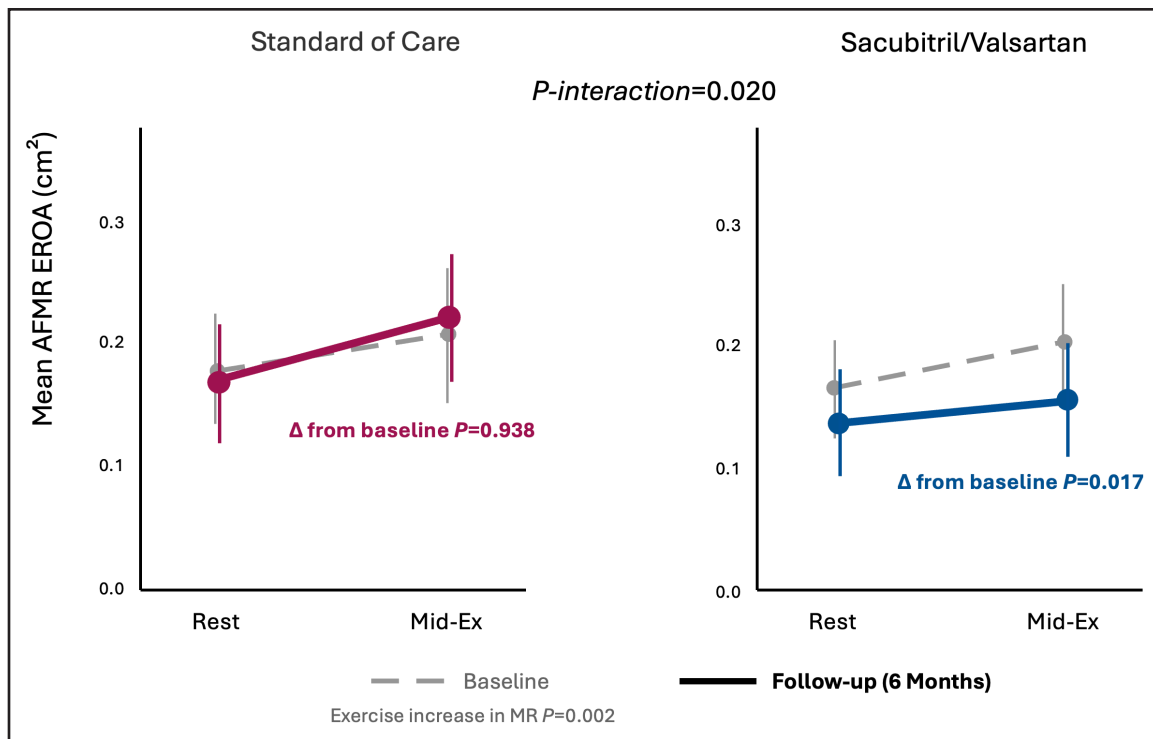


Figure 4. Effect of sacubitril/valsartan on dynamic mitral regurgitation response from rest to mid-exercise.

P values represent the treatment-by-time interaction derived from linear mixed-effect models adjusting for baseline values. 6M indicates 6-month follow-up; EROA, effective regurgitant orifice area; Mid-Ex, mid-exercise; MR, mitral regurgitation; and Δ , delta (change).

exertional AFMR EROA increase ($\geq 5 \text{ mm}^2$) is linked with adverse outcomes.⁸ Whether this results from reverse atrial remodeling or improved mitral leaflet adaptation, as animal models suggest, remains to be determined.^{4,34} Moreover, whether these benefits translate into sustained resting MR reduction over time via chronic remodeling warrants further investigation. Given that the mPAP/CO slope also reflects pulmonary vascular resistance, an additional direct effect on the pulmonary vasculature is also possible especially given the known vasodilatory effects of neprilysin inhibition.³⁵

These improvements in pulmonary pressure–flow relationships parallel outcomes reported with other established left-heart unloading strategies, such as cardiac resynchronization therapy and transcatheter mitral valve repair.^{36,37} Furthermore, the pulmonary pressure-lowering effects of sacubitril/valsartan observed in this study align with real-world evidence from implantable hemodynamic monitoring in broader HFpEF populations, further validating the physiological impact of ARNI on pulmonary hemodynamics.³⁸

The concordance between improvement in patient-reported outcome (KCCQ increase) and exercise capacity (peak VO_2 improvement) provides supporting evidence of the therapy's benefit. Although the open-label nature of the trial warrants a nuanced interpretation of health status scores, it is noteworthy that the observed KCCQ improvement exceeds the approximately 5-point increase reported in placebo arms of contemporary

HFpEF trials.³⁹ In addition, the objective hemodynamic and structural improvements suggest a reduction in disease burden. Furthermore, 60% of patients achieved the target dose, with titration primarily limited by symptomatic hypotension rather than major side effects. This safety profile aligns with robust evidence from broader HFpEF cohorts confirming that sacubitril/valsartan is a generally well-tolerated therapy in this population.^{14,15}

These data underscore that, similar to ventricular functional MR in HFpEF, the management of AFMR within the HFpEF population necessitates optimizing atrio-ventricular myopathy before the true severity of valvular incompetence can be definitively adjudicated.^{18,19} This foundational optimization, encompassing both rhythm management and pharmacological therapy, is essential for accurate risk stratification and determining the potential necessity of valve-specific interventions. These findings underscore that the phenotypic heterogeneity of HFpEF necessitates a more targeted therapeutic approach.

LIMITATIONS

The open-label design of the trial is a notable limitation, as investigators performing the CPETech studies were not blinded to treatment allocation. Although the use of a blinded central core laboratory for the analysis of deidentified images and the requirement to average at least 3 to 5 cardiac cycles per measurement

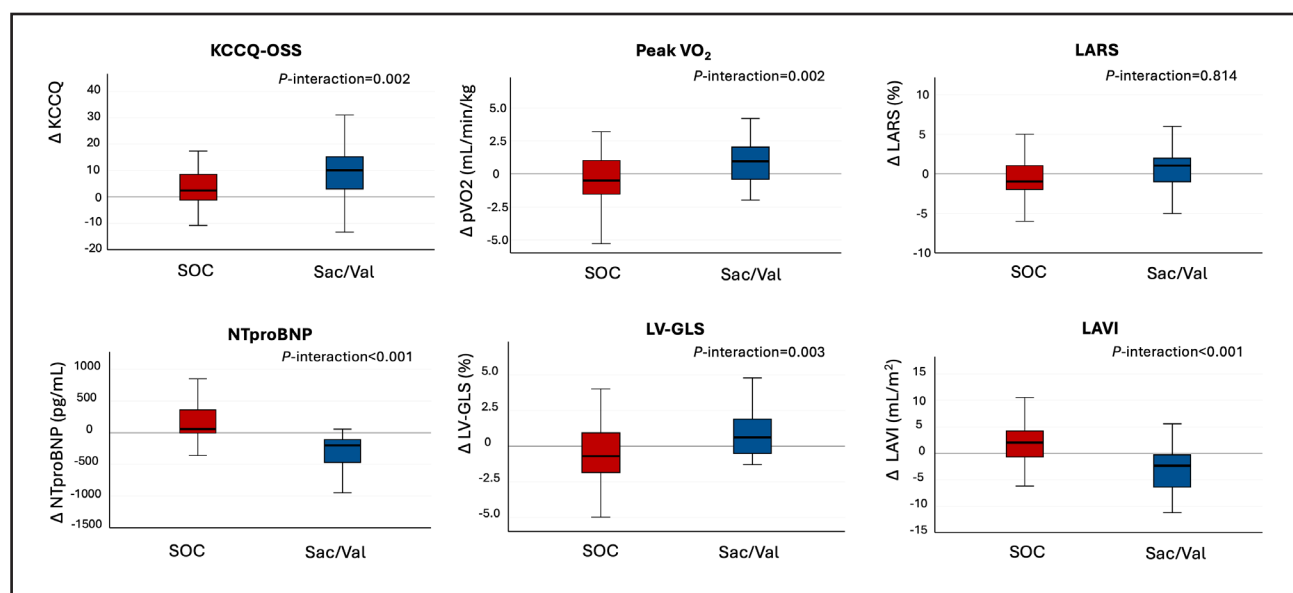


Figure 5. Treatment effects of sacubitril/valsartan on quality of life, exercise capacity, and cardiac structure and function.

P values represent the treatment-by-time interaction derived from linear mixed-effect models adjusting for baseline values. KCCQ-OSS indicates Kansas City Cardiomyopathy Questionnaire Overall Summary Score; LARS, left atrial reservoir strain; LAVI, left atrial volume index; LV-GLS, left ventricular global longitudinal strain; NT-proBNP, N-terminal pro-B-type natriuretic peptide; Peak VO₂, peak oxygen uptake; Sac/Val, sacubitril/valsartan; SOC, standard-of-care; and Δ, delta (change).

were implemented to minimize bias, a double-blinded design would have been superior for eliminating potential investigator influence during image acquisition. This potential for bias should be particularly considered when interpreting subjective endpoints such as quality-of-life scores. Furthermore, although the reliance on noninvasive CPETecho instead of invasive exercise hemodynamics could be viewed as a constraint, this approach enhances the pragmatic utility of the findings for standard practice. The study also utilized surrogate hemodynamic and functional markers over a relatively short 6-month follow-up period; therefore, larger-scale trials are necessary to determine whether these improvements translate into clinical outcomes like mortality or hospitalizations. The achieved sample size was lower than planned (84 versus 110 patients), which may have reduced statistical power. The protocolized in-person follow-up for sacubitril/valsartan titration compared with telephone-based contact for the SOC group may have introduced a degree of performance bias. Finally, the trial was limited to a predominantly Caucasian population, which may limit generalizability to other populations; however, the sex distribution (71% women) was consistent with that typically observed in HFpEF cohorts.

Conclusions

The PRAISE-MR trial shows that sacubitril/valsartan is associated with improvements in exercise hemodynamics, peak VO₂ and health status of patients with HFpEF and AFMR. Treatment was linked with a

blunted increase in dynamic mitral regurgitation during exertion. Although these findings support sacubitril/valsartan for this high-risk phenotype, larger, placebo-controlled outcome trials are warranted to confirm these results.

ARTICLE INFORMATION

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Disclosures

None.

Supplemental Material

CPETecho Protocol and mPAP/CO Calculation
Tables S1–S4
Figures S1–S4
CONSORT Checklist

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